

3MX3 Practice Questions 8

Dr. M. v. Mohrenschildt

z-Transform

Compute the transfer function of $y(n) = x(n) - y(n-1) + x(n-1)$.

What is the frequency response of the system with the transfer function $G(z) = \frac{z}{.5-z}$ (think carefully)

z-Transform

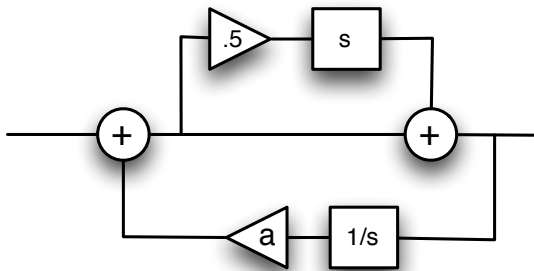
Is $y(n) = x(n) + 2y(n-1) - x(n-1)$ stable ?

Laplace

Determine if $\ddot{y}(t) - 2\dot{y}(t) + y(t) = x(t)$ is stable.

Stability

For which values of the real parameter a is the following system stable ?



Filter

We designed a discrete filter by taking the input, delaying it by 2 samples, multiplying it with .5 and adding it to the input resulting in the output. Show that this system is BIBO stable.